

Yong-chan Park

Ph.D. Candidate, Data Mining Laboratory, Seoul National University | <https://yongchanpark.github.io/>

About me

I am a Ph.D. student majoring in Computer Science and Engineering at [Seoul National University](#) (SNU), advised by [Prof. U Kang](#) in [Data Mining Laboratory](#). My research interests include machine learning, tensor analysis, and anomaly detection.

Research vision and current work

My research lies at the intersection of machine learning and large-scale tensor analysis. I develop algorithms that make high-dimensional data representations both *faster* and *more interpretable*, with particular emphasis on streaming and irregular tensors common in web-scale applications. Recent projects include

- **Partial Fourier Transform:** a linear-time Fourier transform that accelerates spectral anomaly detection by $\times 20$ over prior art.
- **Data-Axis Transformer with Multi-Level Contexts:** the first end-to-end architecture that models inter-stock dependencies across markets for price-movement prediction.
- **PuzzleTensor:** a method-agnostic transformation that reduces the target rank of arbitrary tensors while preserving reconstruction fidelity.

Education

Ph.D. Student

- Computer Science and Engineering, SNU

Mar. 2020 - Present

Bachelor of Science

- Department of Mathematical Sciences, SNU

Mar. 2012 - Feb. 2019

Publications

- [1] **Yong-chan Park**, Jun-Gi Jang, and U Kang. "Fast and Accurate Partial Fourier Transform for Time Series Data." KDD 2021
- [2] Jaemin Yoo, Yejun Soun, **Yong-chan Park**, and U Kang. "Accurate Multivariate Stock Movement Prediction via Data-Axis Transformer with Multi-Level Contexts." KDD 2021
- [3] **Yong-chan Park**^{*}, Sangjun Son^{*}, Minyong Cho, and U Kang. "DAO-CP: Data-Adaptive Online CP Decomposition for Tensor Stream." PLOS One 2022 (*equal contribution)
- [4] Jun-Gi Jang, Jeongyoung Lee, **Yong-chan Park**, and U Kang. "Fast and Accurate Dual-Way Streaming PARAFAC2 for Irregular Tensors - Algorithm and Application." KDD 2023
- [5] **Yong-chan Park**, Jongjin Kim, and U Kang. "Fast Multidimensional Partial Fourier Transform with Automatic Hyperparameter Selection." KDD 2024
- [6] Jun-Gi Jang, **Yong-chan Park**, and U Kang. "Fast and Accurate PARAFAC2 Decomposition for Time Range Queries on Irregular Tensors." CIKM 2024
- [7] JinGee Kim, **Yong-chan Park**, Jaemin Hong, and U Kang. "Accurate Stock Movement Prediction via Multi-Scale and Multi-Domain Modeling." BigData 2024
- [8] SeungJoo Lee, **Yong-chan Park**, and U Kang. "Accurate Coupled Tensor Factorization with Knowledge Graph." BigData 2024
- [9] **Yong-chan Park**, Kisoo Kim, and U Kang. "PuzzleTensor: A Method-Agnostic Data Transformation for Compact Tensor Factorization." KDD 2025

Under review

[R1] **Yong-chan Park**, SeungJoo Lee, and U Kang. “Fast and Accurate Online Coupled Matrix-Tensor Factorization via Frequency Regularization.”

[R2] **Yong-chan Park**, SeungJoo Lee, and U Kang. “CoCoNat: Leveraging Cross-Query Context to Enhance Named Entity Recognition.”